

Experiment 2

Problem Identification and Feasibility: Software Design Principles

Aim

To identify the problem addressed by the LMS Platform and prepare its problem statement, objectives, scope, and technical and operational feasibility analysis.

Objectives

- Identify the real-world problem addressed by the software project.
- Prepare a clear problem statement.
- Define project objectives and scope.
- Perform technical and operational feasibility analysis.
- Evaluate the viability of the proposed solution.

Introduction

Problem identification is the first step in software engineering. A well-defined problem statement and feasibility study ensure that the proposed software solution is practical, achievable and aligned with stakeholder requirements before development begins.

Selected Project

Project Title: LMS Platform

Domain: Education Technology and Web Application Development

Problem Statement

Students often struggle to access structured, reliable and affordable learning resources from a single platform. Free educational content is scattered across multiple sources, premium coaching and study material can be expensive, and students have limited ways to attempt structured tests, track their performance or attend live classes conducted by qualified educators. Educators, in turn, lack a unified system to upload content, conduct live classes, create tests and monitor student progress, while institutions need a way to manage users, content, payments and analytics from a single dashboard. The LMS Platform addresses this problem by providing a centralized, role-based learning management system that combines free and paid educational content, online test series, live and recorded classes, secure payments and performance analytics for students, educators and administrators.

Project Objectives

- Provide a centralized online learning environment for students, educators and administrators.
- Make educational resources easily accessible through free and paid content.
- Provide an online test and assessment system with automatic evaluation.
- Enable educators to conduct live classes and upload recorded lectures.
- Implement secure online payments and subscription management.
- Provide analytics and performance tracking for students, educators and administrators.
- Build a scalable and modular educational platform architecture.

Project Scope

The LMS Platform supports:

- Informational public website and role-based authentication.
- Course management with free and paid notes.
- Online test series with MCQs, timers, scoring and leaderboards.
- Live classes with scheduling, attendance and live Q&A.
- Recorded lecture hosting with subscription/purchase-based access.
- Payment and subscription system (monthly, yearly and one-time purchases).
- Dedicated dashboards for students, educators and administrators.
- Content management system for notes, blogs and news updates.
- Analytics and reporting for students, educators and administrators.

Features such as native mobile applications, offline content access and advanced DRM implementation are outside the scope of the initial (Phase 1) version of this project.

Technical Feasibility Analysis

Factor	Assessment	Remarks
Technology	Feasible	Next.js/React.js, Node.js, REST APIs and PostgreSQL/MongoDB can support development.
Hardware	Feasible	Standard computers and smartphones with internet access and a camera/microphone are sufficient.
Software Tools	Feasible	IDEs, Git, GitHub, Docker, cloud services and web-development frameworks are readily available.
Security	Feasible	JWT-based authentication, role-based authorization and secure payment processing can be implemented.
Scalability	Feasible	Cloud-based and modular architecture can support increasing users, courses and video traffic.
Media & Payments	Feasible	CDN-based video delivery (AWS S3/Cloudflare R2) and payment gateways (Razorpay/Stripe) can be integrated.

Operational Feasibility Analysis

Factor	Assessment	Remarks
User Acceptance	High	Students, educators and administrators can use a familiar dashboard-based interface.
Ease of Use	High	The platform provides guided dashboards, clear navigation and structured content organization.
Availability	High	Students can access content, tests and recorded lectures anytime without depending on fixed schedules.
Maintainability	High	Modular design allows student, educator, admin, payment and content modules to be maintained independently.

Factor	Assessment	Remarks
Business Benefit	High	Reduces dependence on scattered resources and enables monetization through subscriptions and course sales.
Performance Improvement	High	Test analytics and performance tracking help students and educators identify and address weak areas.

Analysis

The feasibility study indicates that the LMS Platform is technically and operationally viable. Existing web-development, database, cloud storage, payment gateway and video-delivery technologies can support the implementation of the proposed system. The platform can provide students with accessible and structured learning resources while giving educators and administrators the tools needed to manage content, classes, tests and payments effectively.

Observation

Clearly defining the problem, objectives, scope and feasibility of the LMS Platform provides a strong foundation for its subsequent software design, development and implementation.

Result

The problem statement, objectives, project scope, technical feasibility analysis and operational feasibility analysis for the LMS Platform were successfully prepared.